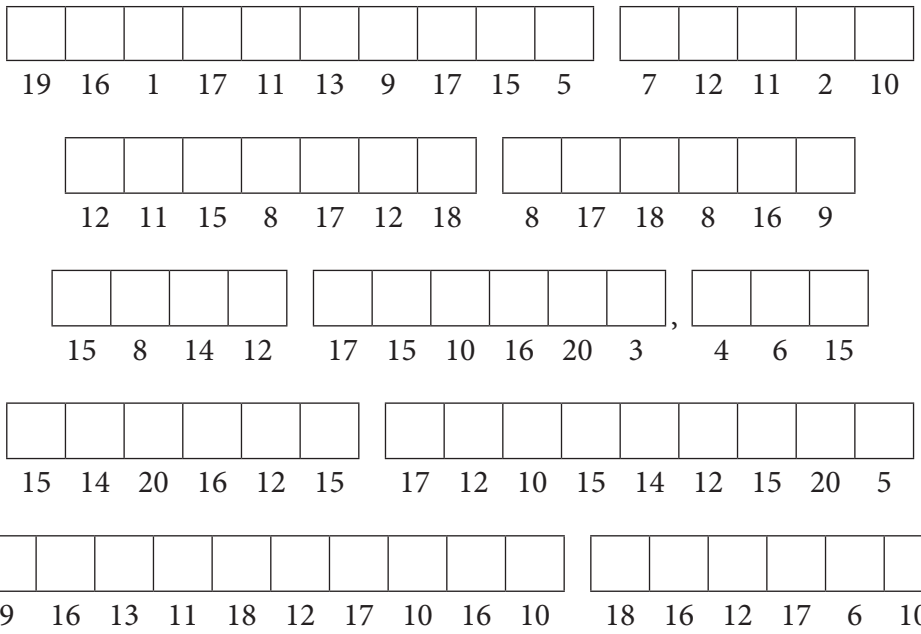
Name: _____

WORKSHEET

Fractional indices and radicals



Sir Arthur Conan Doyle

Rule

$a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$, where a $>$ 0 and n $>$ 0.

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Questions

Express the following radicals as powers.

1 $\sqrt[3]{x^5}$

2 $\sqrt[2]{y^7}$

3 $\sqrt[5]{x}$

4 $\sqrt{y^{11}}$

Express the following powers as radicals.

5 $x^{\frac{2}{3}}$

6 $y^{\frac{3}{4}}$

7 $x^{\frac{2}{5}}$

8 $y^{\frac{1}{2}}$

Evaluate the following.

9 $49^{\frac{1}{2}}$

10 $16^{\frac{3}{4}}$

11 $\left(8^{\frac{2}{7}}\right)^0$

12 $125^{\frac{2}{3}}$

Simplify the following and give your answers as powers.

13 $2x^{\frac{1}{2}} \times 3x^{\frac{1}{4}} \times x^2 =$

14 $10y^{\frac{5}{6}} \div 5y^{\frac{2}{3}} \times 4y^{\frac{1}{2}} =$

15 $3\sqrt[5]{x^2} \times 2x^{\frac{3}{10}} \div \sqrt{x} =$

16 $\dfrac{14\sqrt[5]{y^2}}{y^{\frac{3}{4}}} \times \dfrac{y^{\frac{1}{4}}}{21\sqrt[3]{y}} =$

Simplify the following and give your answers as radicals.

17 $\sqrt{x} \times \sqrt[5]{x^2} =$

18 $\sqrt[4]{y^3} \div \sqrt[3]{y^2} =$

19 $15x^{\frac{3}{4}} \div 5\sqrt[10]{x^3} \times 2x^{\frac{2}{5}} =$

20 $\dfrac{\sqrt{y^5}}{6y^{\frac{4}{3}}} \times \dfrac{18\sqrt[6]{y^2}}{y^{\frac{3}{5}}} =$

Solutions

A $8y^{\frac{2}{3}}$

B $y^{\frac{11}{2}}$

C $6x^{2\frac{3}{4}}$

D $x^{\frac{5}{3}}$

E $\dfrac{2}{3y^{\frac{13}{30}}}$

F $x^{\frac{1}{5}}$

G $\sqrt[12]{y}$

H $\sqrt{y}$

I $\sqrt[10]{x^9}$

K $\sqrt[5]{x^2}$

L $3\sqrt[10]{y^9}$

M $6\sqrt[20]{x^{17}}$

N 25

O 1

R 7

S 8

T $6x^{\frac{1}{5}}$

U $\sqrt[4]{y^3}$

W $y^{\frac{7}{2}}$

Y $\sqrt[3]{x^2}$

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